BUSINESS PROBLEM STATEMENT

Medical Data History

1. Business Problem Statement

Healthcare organizations maintain patient and hospital admission records to support day-to-day operations and the review of patient activity. These records contain important information about patients' demographic details, allergies, physical measurements, locations, admission and discharge dates, diagnoses, and the doctors responsible for their care. When this information is stored across related tables, retrieving a clear and reliable view requires structured data querying.

The Medical Data History project uses SQL to examine the provided relational database and answer a defined set of 34 problem queries. The analysis involves filtering patient records, handling missing allergy values, summarizing admissions, identifying recurring diagnoses, comparing patient characteristics, and combining information from related tables. The goal is to convert the available records into accurate, well-organized query results that address each business question and demonstrate how SQL can support the analysis of medical and hospital data.

2. Project Analysis Tasks

1. Show the first name, last name, and gender of patients whose gender is 'M'.

2. Show the first name and last name of patients who do not have allergies.

3. Show the first names of patients that start with the letter 'C'.

4. Show the first name and last name of patients whose weight is within the range of 100 to 120 (inclusive).

5. Update the allergies column in the patients table. If a patient's allergies value is NULL, replace it with 'NKA'.

6. Show the first name and last name concatenated into one column to display the patient's full name.

7. Show the first name, last name, and full province name of each patient.

8. Show how many patients have a birth_date with 2010 as the birth year.

9. Show the first_name, last_name, and height of the patient with the greatest height.

10. Show all columns for patients with patient_id values 1, 45, 534, 879, and 1000.

11. Show the total number of admissions.

12. Show all columns from admissions where the patient was admitted and discharged on the same day.

13. Show the total number of admissions for patient_id 579.

14. Based on the cities where patients live, show the unique cities that are in province_id 'NS'.

15. Find the first_name, last_name, and birth date of patients whose height is more than 160 and weight is more than 70.

2. Project Analysis Tasks (Continued)

16. Show unique birth years from patients and order them in ascending order.

17. Show unique first names from the patients table that occur only once in the list. Exclude names that appear more than once.

18. Show patient_id and first_name for patients whose first_name starts and ends with 's' and is at least 6 characters long.

19. Show patient_id, first_name, and last_name for patients whose diagnosis is 'Dementia'. The primary diagnosis is stored in admissions.

20. Display every patient's first_name. Order the list by the length of each name and then alphabetically.

21. Show the total number of male patients and the total number of female patients in the patients table. Display both results in the same row.

22. Show the total number of male patients and the total number of female patients in the patients table. Display both results in the same row.

23. Show patient_id and diagnosis from admissions. Find patients admitted multiple times for the same diagnosis.

24. Show the city and the total number of patients in each city. Order from the most patients to the least, then by city name ascending.

25. Show the first name, last name, and role of every person who is either a patient or a doctor. The roles are 'Patient' or 'Doctor'.

26. Show all allergies ordered by popularity. Remove NULL values from the query.

27. Show each patient's first_name, last_name, and birth_date for patients born in the 1970s. Sort from the earliest birth_date.

28. Display each patient's full name in one column: last_name in uppercase first, then first_name in lowercase, separated by a comma. Order by first_name descending. Example: SMITH,jane.

29. Show province_id and the sum of height for provinces where the total sum of patient height is greater than or equal to 7,000.

30. Show the difference between the largest weight and smallest weight for patients with the last name 'Maroni'.

31. Show all days of the month (1–31) and how many admission_dates occurred on each day. Sort from the most admissions to the fewest.

32. Group all patients into weight groups and show the total number of patients in each group. Order weight groups descending. For example, weights 100–109 belong to group 100; 110–119 belong to group 110.

33. Show patient_id, weight, height, and isObese from patients. Display isObese as a Boolean 0 or 1. Obesity is defined as weight (kg) / height (m); height is stored in cm.

34. Show patient_id, first_name, last_name, and the attending doctor's specialty. Include only patients with diagnosis 'Epilepsy' whose doctor's first name is 'Lisa'. Use patients, admissions, and doctors.

3. Dataset Description

The Medical Data History project uses the relational database **project_medical_data_history**. The database is organized into four connected tables that represent patient details, hospital admissions, doctors, and province information. Together, these tables provide the fields required to investigate patient demographics, health-related attributes recorded in the dataset, admission activity, diagnoses, and the doctors associated with patient admissions. The project questions use these fields to produce individual results as well as grouped and combined summaries.

4. Attribute Information

Table	Attribute	Description
patients	patient_id	Unique identifier assigned to each patient.
	first_name, last_name	Patient's first and last names.
	gender	Patient's gender.
	birth_date	Patient's date of birth.
	city, province_id	City where the patient lives and the associated province identifier.
	allergies	Allergy information recorded for the patient.
	height, weight	Patient's height in centimeters and weight in kilograms.
admissions	patient_id	Identifier of the patient associated with the admission.
	admission_date, discharge_date	Dates on which the patient was admitted and discharged.
	diagnosis	Diagnosis recorded for the admission.
	attending_doctor_id	Identifier of the doctor attending the admission.
doctors	doctor_id	Unique identifier assigned to each doctor.
	first_name, last_name	Doctor's first and last names.
	specialty	Medical specialty of the doctor.
province_names	province_id	Identifier assigned to a province.
	province_name	Full name of the province.

5. Project Deliverables

The project requires the 34 problem queries in the supplied statement to be solved using SQL. The work involves writing and executing queries to retrieve the requested patient and admission information, apply the specified conditions and calculations, summarize records, and combine related data where required. The completed SQL queries and their corresponding results form the core outputs of the analysis.